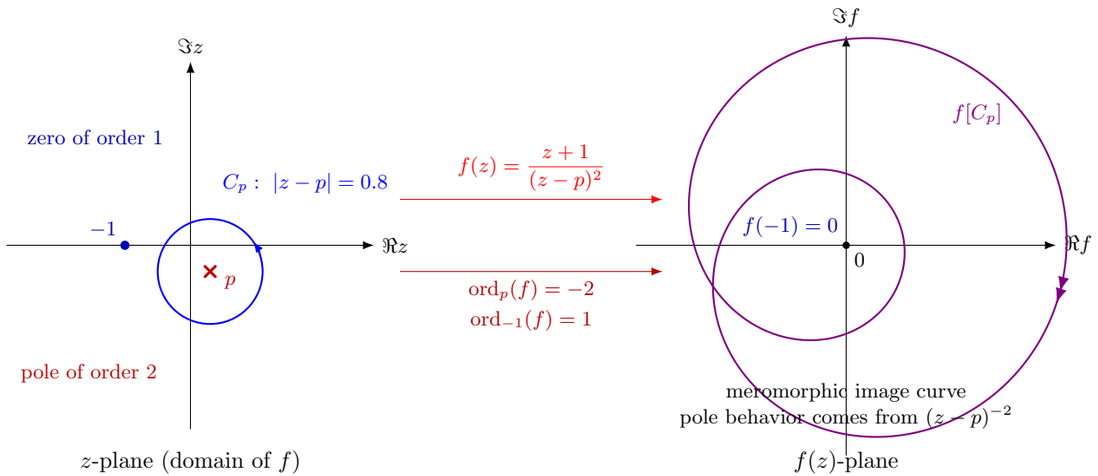
$$D = 2p, \qquad L(D) = L(2p) = \{f \text{ meromorphic } : \operatorname{div}(f) + 2p \geq 0\}.$$

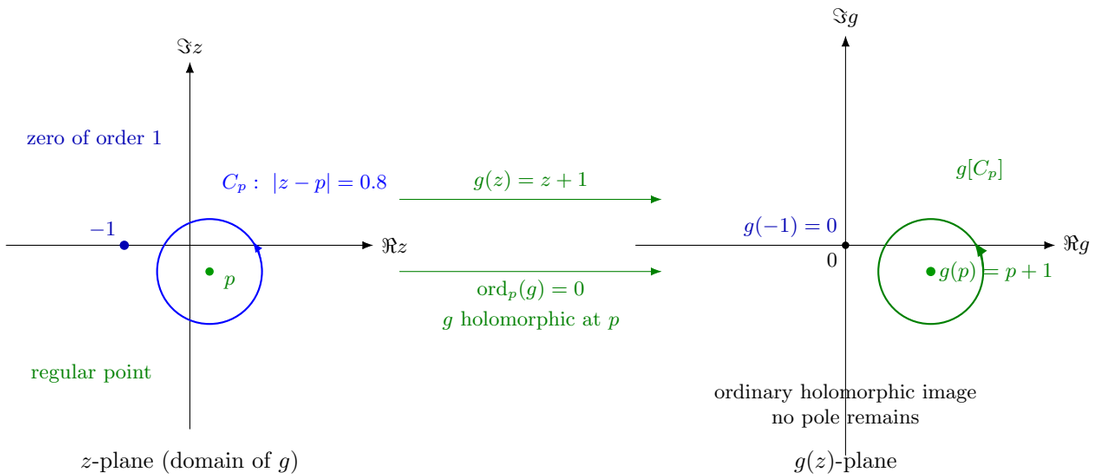
Equivalently,

$$f \in L(2p) \iff \operatorname{ord}_p(f) + 2 \geq 0 \iff (z-p)^2 f \text{ is holomorphic at } p.$$



correct by $(z-p)^2$

$g(z) := (z-p)^2 f(z) = z+1$



$$\operatorname{ord}_p(f) + 2 = -2 + 2 = 0 \geq 0.$$

Hence

$$f \in L(2p).$$

Geometrically: the divisor $D = 2p$ allows exactly enough pole at p so that

$$(z-p)^2 f$$

becomes holomorphic.

For contrast, if

$$h(z) = \frac{1}{(z-p)^3},$$

then

$$\operatorname{ord}_p(h) + 2 = -3 + 2 = -1 < 0,$$

so

$$h \notin L(2p).$$

Indeed,

$$(z-p)^2 h = \frac{1}{z-p}$$

still has a pole at p.